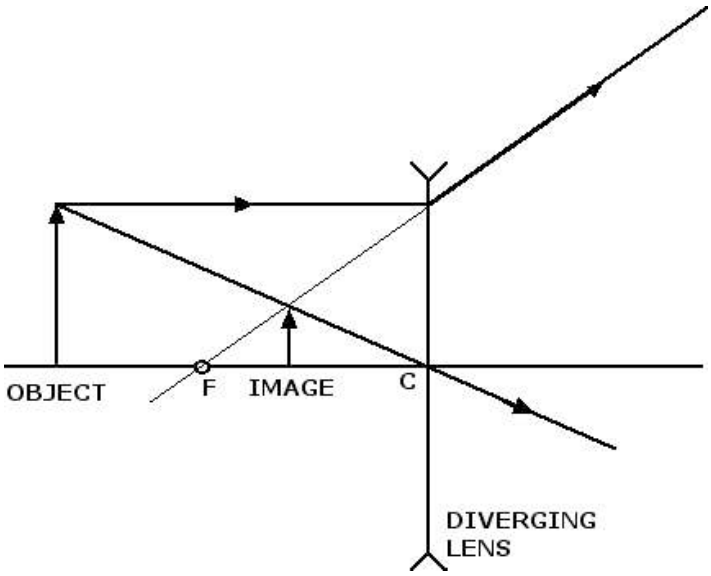
the focal point of the lens) and the ray which passes through the optical center (which is not deviated by the lens). Like in the case of convex mirrors, the image is always virtual, reduced in size and upright, as it is clear from the following picture:



We have just seen that the image formation can be done case by case using the refraction properties of some particular rays of light. With this procedure we can find out the position of the images w.r.t. to the lens. We are also able to foresee such a position by using a suitable law which is called *conjugated point formula*. If we indicate with p the distance between the object and the optical center, with f the focal length and with q the distance between the image and the optical center we have that $\frac{1}{p} + \frac{1}{q} = \frac{1}{f}$. This formula is identical to the one which holds for mirrors. Nevertheless there are some differences in the conventions about the signs. In our case the distance q is positive when the image is beyond the lens (w.r.t. to the object and the incident rays), the distance q is negative when the image is in front of the lens, i.e. the object and the image are both to the left or to the right of the lens.

As a consequence of this convention, we have that in a convergent lens $q > 0$ when the image is real and inverted [cases 1) and 2)], $q < 0$ when the image is virtual and upright [case 3)]. In a divergent lens instead q is always negative, because the image is always in front of the lens. For what concerns the focal length, it is positive $f > 0$ if the lens is convergent, it is negative $f < 0$ if the lens is divergent.

The height of the image h_i can be obtained from the height of the object h_o using the formula $h_i = I \cdot h_o$. If the magnification $I > 1$ then the image is enlarged, if $I < 1$ then the image is reduced in size w.r.t. the object. It is easy to prove that the magnification I can be obtained from the distance of the image q and of the object p using the formula $I = \frac{|q|}{p}$.

To conclude this section, we can define the diopter power d of the lens: it is the inverse of the focal length $d = \frac{1}{f}$. If we measure f in meters, the ratio $\frac{1}{f}$ gives us the number of diopters of the lens. For example, the focal length of a 2 diopter lens is $1/2\ m = 0.5\ m$.